

# Inline and Display Mathematics

Source-backed grouped formula sample

## 1 Expressions

**Expression 1.** The following expression is taken from a source-backed formula corpus.

$$\mathcal{I}_D(\delta) = \sup_{\gamma \in \Sigma_D(\delta)} \int_{\mathcal{Y}_1 \times \mathcal{Y}_2} f(y_1, y_2) d\gamma(y_1, y_2) \mathcal{I}(\delta) = \sup_{\gamma \in \Sigma(\delta)} \int_{\mathcal{S}} f(y_1, y_2) d\gamma(y_1, y_2, x).$$

**Expression 2.** The following expression is taken from a source-backed formula corpus.

$$\mathcal{I}(\delta_1, 0) = \widehat{\mathcal{I}}(\delta_1, \delta_2) = \inf_{\lambda_1 \in \mathbb{R}_+} \left[ \langle \lambda, \delta \rangle + \sup_{\varpi \in \Pi(\mu_{13}, \mu_{23})} \int_{\mathcal{V}} f_{\lambda}(s_1, s_2) d\varpi(s_1, s_2) \right],$$

**Expression 3.** The following expression is taken from a source-backed formula corpus.

$$y^{\delta} \tilde{\Gamma}^{\alpha}_{\beta\delta} \tilde{B}^{\beta}|_{1+3} = y^{\delta} dx^{\sigma} [\Gamma^{\alpha}_{\beta\delta} B^{\beta}_{,\sigma} + \Gamma^{\alpha}_{\beta\delta} \Gamma^{\beta}_{\sigma\tau} B^{\tau} + \Gamma^{\alpha}_{\beta\delta,\sigma} B^{\beta}].$$

**Expression 4.** The following expression is taken from a source-backed formula corpus.

$$\theta_{\perp\parallel} = \frac{1}{y^2} (\partial_{\tau} \vec{x} \partial_{\sigma} \vec{x} + \partial_{\tau} y \partial_{\sigma} y) = \frac{1}{a^2 \tau} \left[ \vec{f} \vec{c}' + aa' \right] + \frac{1}{a^2} (\vec{g} \vec{c}') + \dots$$

**Expression 5.** The following expression is taken from a source-backed formula corpus.

$$\left. \frac{d}{d\tau} \right|_{\tau=0} x^{\alpha}(g_A(\tau)g) = \left. \frac{\partial x^{\alpha}(g_A(\tau)g)}{\partial x^M(g_A(\tau))} \right|_{g_A=e} \cdot \left. \frac{dx^M(g_A(\tau))}{d\tau} \right|_{\tau=0} = R^{\alpha}_M(g) \delta^M_A = R^{\alpha}_A(g).$$